

An Explainable Optimised Stacking Ensemble Framework for Heart Attack Prediction: Reproduction and Enhancement of a Machine Learning Approach

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Assessment: Task 11.1HD

Research Background & Objectives

Background:

- Cardiovascular disease is a major global health challenge.
- Machine learning can support early heart disease prediction.
- Ensemble models provide strong predictive performance.

Objectives:

- Reproduce a published stacking ensemble framework.
- Critically analyse its reliability.
- Develop an improved explainable stacking framework.



Reproduction of Published Research

Selected Research Paper: An efficient stacking-based ensemble technique for early heart attack prediction(Bhagat et al., 2025)

Original Framework:

Dataset:

- Heart disease dataset
- 1,025 patient records
- 13 clinical features

Base Machine Learning Models:

- Logistic Regression
- Decision Tree
- Random Forest
- XGBoost
- Naive Bayes
- K-Nearest Neighbours

Reproduction Objective:

- Implement the published stacking ensemble framework.
- Evaluate whether the reported performance can be reproduced.
- Analyse dataset quality and model reliability.

6 ML Classifiers

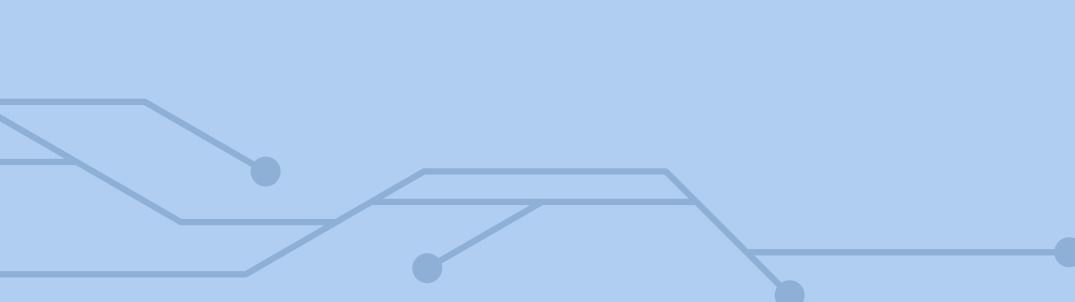


Stacking Ensemble



Heart Attack Prediction





Dataset Quality Analysis and Reproduction Findings

Dataset Investigation:

Heart Disease Dataset:

- 1,025 Records
- 13 Clinical Features

Duplicate Records

Identified:

- 723 Duplicate Samples

Clean Dataset:

- 302 Unique Patient Records

Reproduction Result:

Original Dataset Accuracy: 100%



After Duplicate Removal

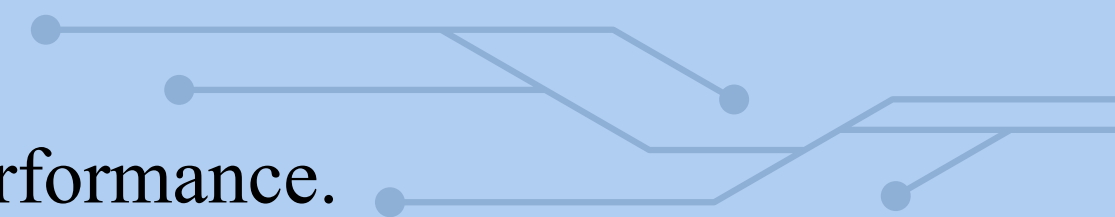
More realistic evaluation



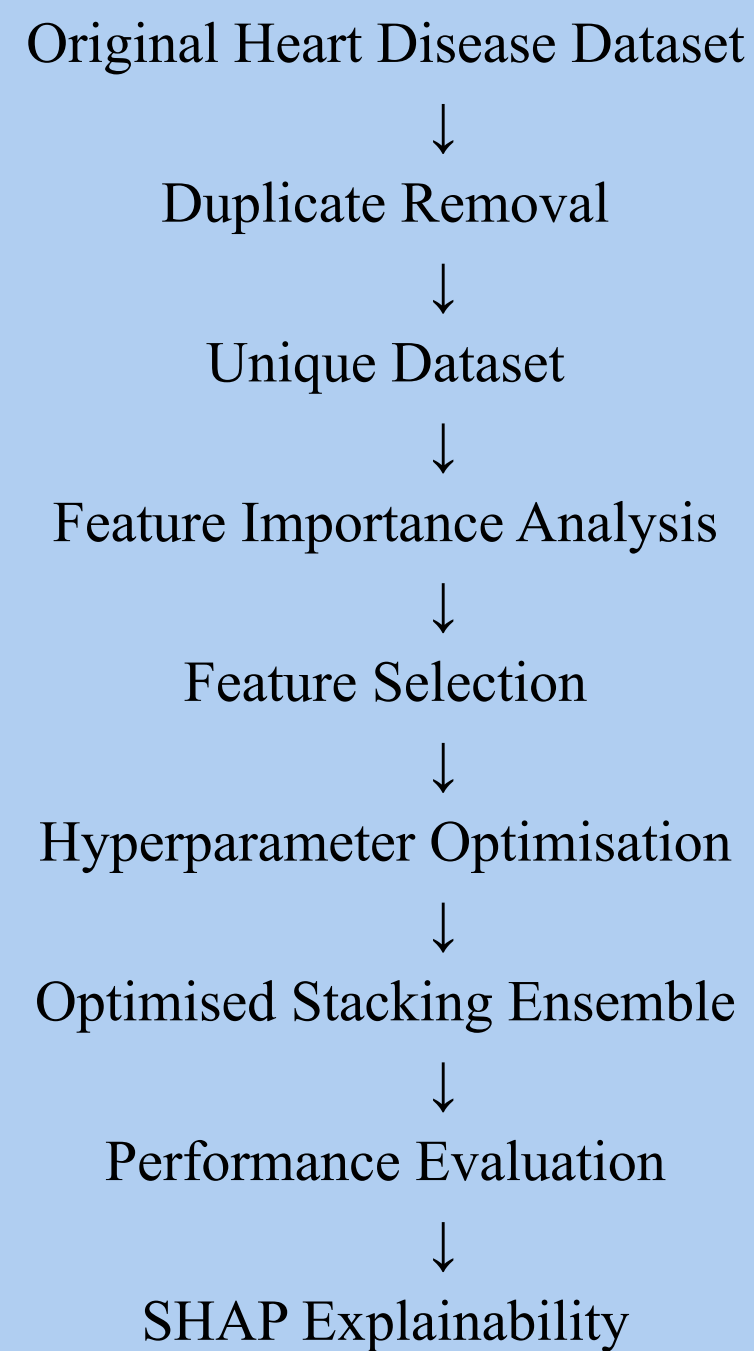
Performance decreased

Key Finding:

Duplicate records can lead to overestimated machine learning performance.



Proposed Optimised Stacking Ensemble Framework



Key Improvements Over Original Approach:

- Data quality validation
- Feature-based optimisation
- Hyperparameter tuning
- Explainable AI integration

Original Approach	Proposed Framework
Stacking ensemble	Optimised stacking ensemble
Original dataset	Duplicate-free dataset
Performance focused	Performance + reliability + explainability
No SHAP analysis	SHAP-based interpretation

Main Contribution:

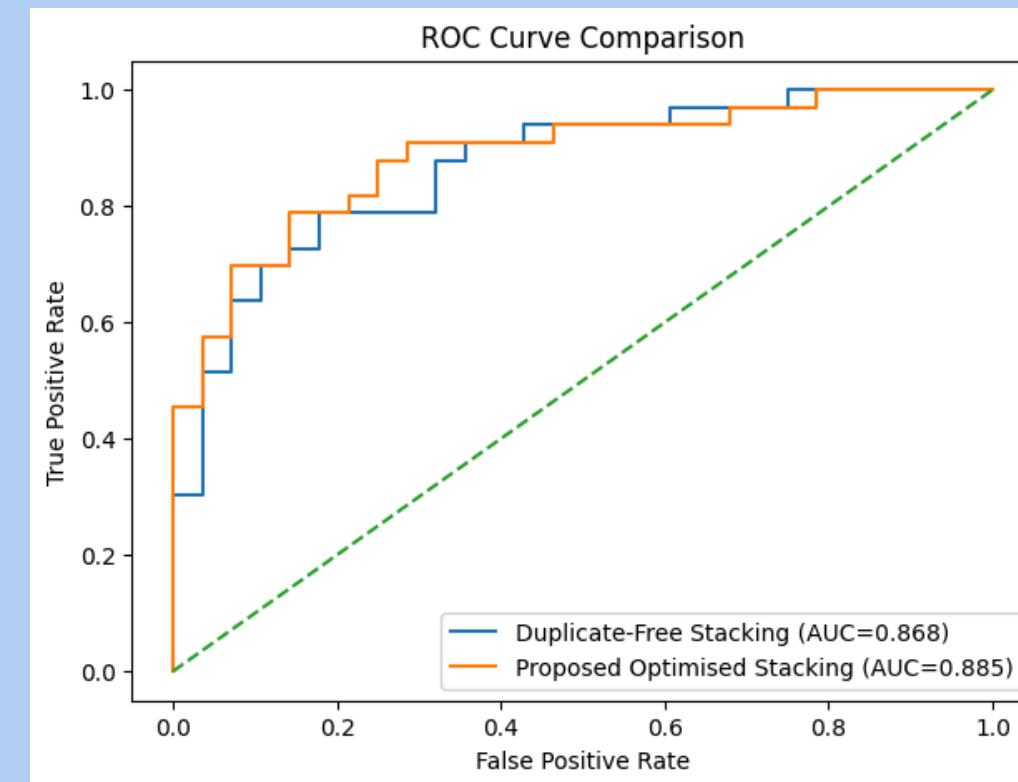
A reliable and interpretable stacking framework evaluated on a duplicate-free dataset.

Experimental Results and Performance Evaluation

Model	Accuracy	F1-score	AUC
Duplicate-Free Stacking Baseline	75.41%	77.61%	0.868
Proposed Optimised Stacking	81.97%	84.06%	0.885

Improvement Achieved:

- Accuracy improvement: +6.56%
- F1-score improvement: +6.45%
- AUC improvement: 0.868 → 0.885

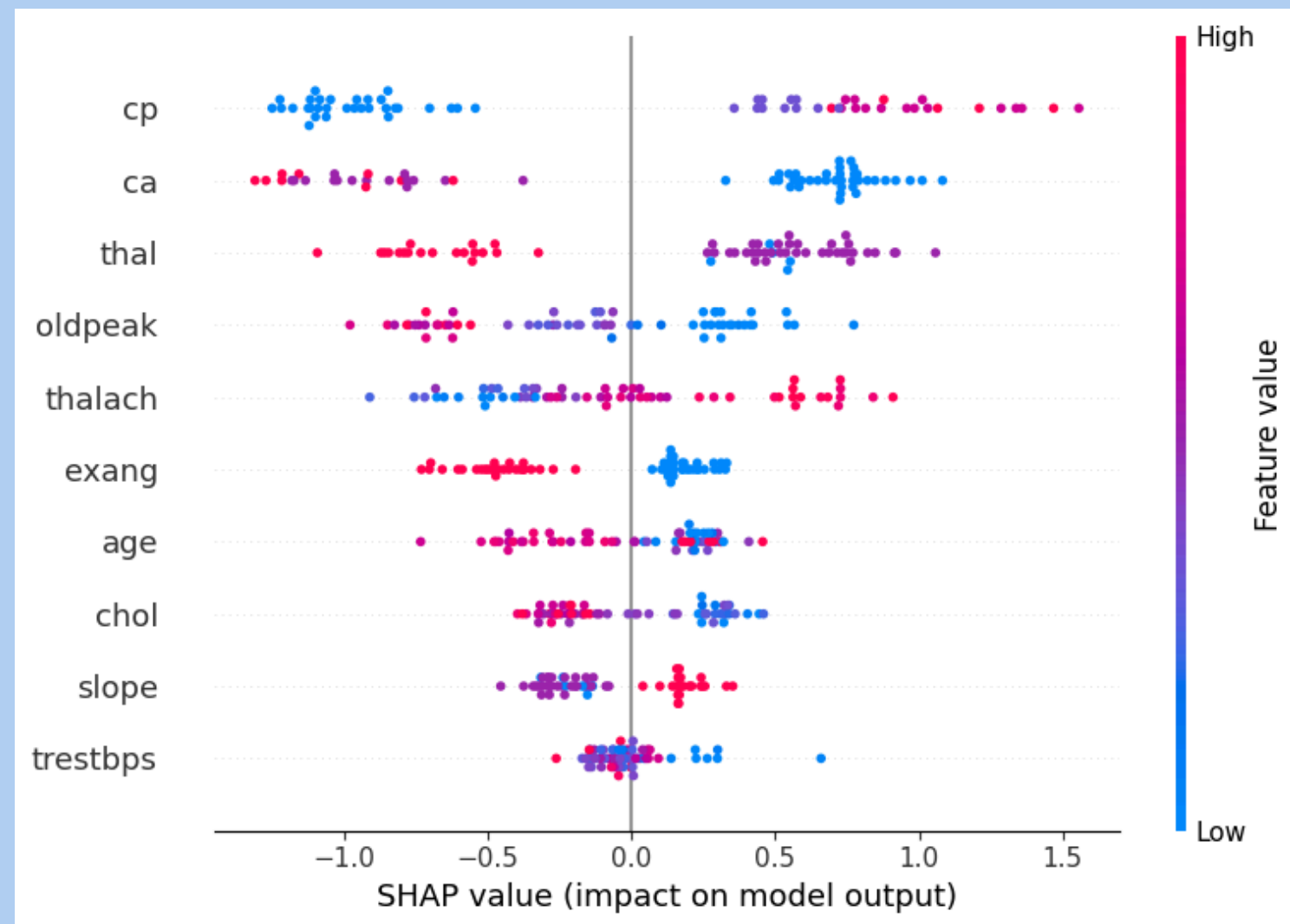


Key Result:

The proposed framework improved predictive performance while maintaining evaluation on a duplicate-free dataset.

Explainable AI Analysis Using SHAP

SHAP Summary Plot: Feature Contributions:



Key Influential Features:

- cp — Chest pain type
- ca — Number of major vessels
- thal — Thalassemia indicator
- oldpeak — ST depression induced by exercise
- thalach — Maximum heart rate achieved

Key Finding:

SHAP analysis improves model transparency by explaining how clinical features influence heart attack predictions.

Conclusion and Key Contributions



Key Contributions:

Reproduced a published stacking ensemble framework

- Implemented the original heart attack prediction approach.

Identified dataset reliability issues

- Detected duplicate records affecting model evaluation.

Developed an optimised stacking framework

- Integrated data validation, feature selection, and hyperparameter optimisation.

Improved predictive performance

- Achieved better results on a duplicate-free dataset.

Integrated SHAP explainability

- Provided interpretable insights into clinical feature contributions.

Reliable healthcare machine learning requires not only high predictive performance, but also trustworthy data evaluation and model interpretability.

Thank You